BioPhorum

CELL THERAPY PROCESS MANUFACTURING MAPS

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Contents

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About BioPhorum

BioPhorum's mission is to create environments where the global biopharmaceutical industry can collaborate and accelerate its rate of progress, for the benefit of all.

Since its inception in 2004, BioPhorum has become the open and trusted environment where senior leaders of the biopharmaceutical industry come together to openly share and discuss the emerging trends and challenges facing their industry.

Growing from an end-user group in 2008, BioPhorum now comprises over 90 manufacturers and suppliers deploying their top 3,500 leaders and subject matter experts to work in seven focused Phorums, articulating the industry's technology roadmap, defining the supply partner practices of the future, and developing and adopting best practices in drug substance, fill finish, process development and manufacturing IT. In each of these Phorums, BioPhorum facilitators bring leaders together to create future visions, mobilize teams of experts on the opportunities, create partnerships that enable change and provide the quickest route to implementation, so that the industry shares, learns and builds the best solutions together.

1.0

Introduction

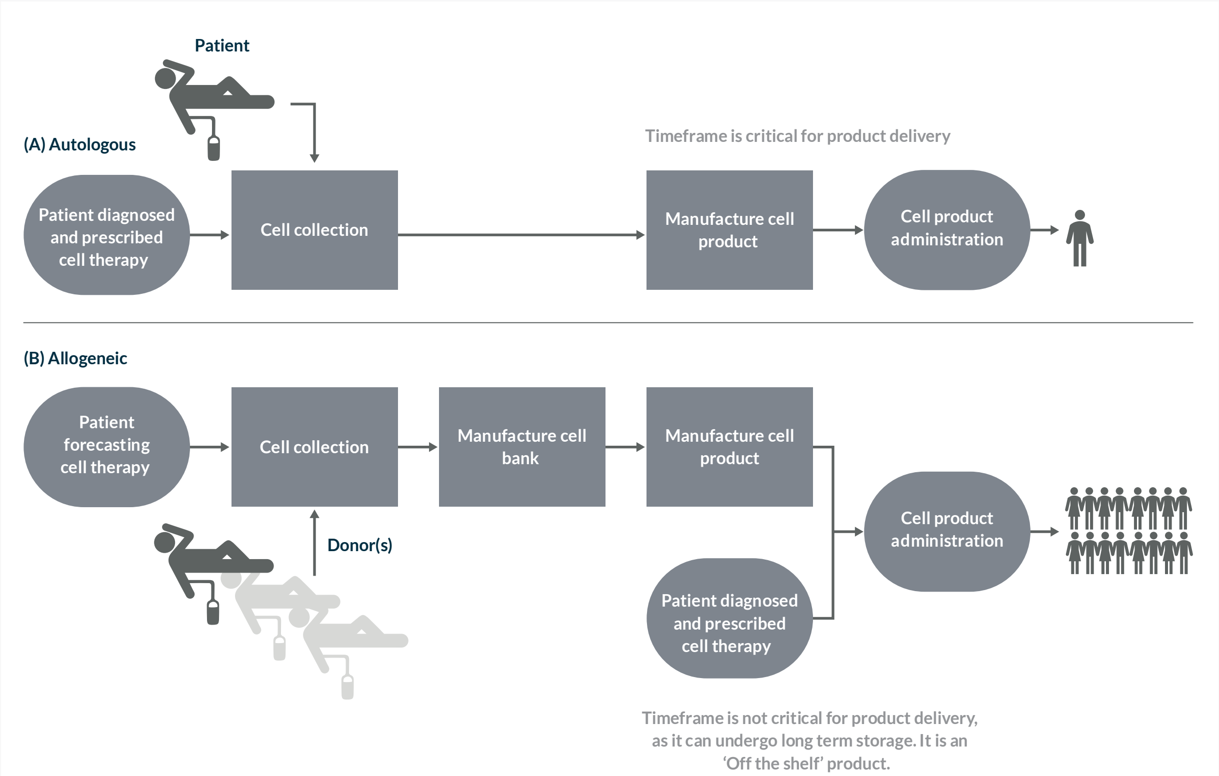
Until recently, there were limited options for patients seeking treatment for ailments or diseases relating, but not limited to, cancer, inflammatory diseases, tissue replacement and blood disorders. However, over this time, a new therapeutic pillar began to emerge through the support of academic centers. Today, the academic vision of shifting treatment paradigms has fully evolved to clinically proven results and market approval – all of this is in thanks to cell and gene-modified cell therapies.

Recent industry reports show that there are over 630 clinical trials (and growing) underway for cell and gene-modified cell therapies with nearly 55% of these in Phase II*. In addition, many companies apply for the Regenerative Medicine Advanced Therapy (RMAT) designation to speed up the development process. The numbers are staggering and these therapies pose unique manufacturing challenges. Unlike the conventional biologics industry with the ability to leverage templated manufacturing processes, cell and gene-modified cell therapy production standards have yet to be established.

Here, BioPhorum's Cell & Gene Therapy Commercialization workstream has generated a set of cell and gene-modified cell therapy process maps that represent example processes, unit operations and platform technologies being used in cell and gene therapy manufacturing today. A high-level overview of autologous and allogeneic cell therapy manufacturing is shown in Figure 1. As the focus of this article is cell therapy, manufacturing details for viral gene editing agents (such as lentivirus), are not shown. Viral vector manufacturing and other processes to support *in vivo* and *ex vivo* gene therapies are being mapped by another team in the BioPhorum Cell & Gene Therapy Commercialization workstream www.biophorum.com/download/gene-therapy-process-map/

**Reference: Alliance for Regenerative Medicine, Q3 2019 Report*

Figure 1: Autologous and allogeneic cell therapy manufacture overview



The intent of the cell therapy process maps provided in the appendices is to reflect the current state of selected manufacturing processes within the cell therapy field, form a common frame of reference for scientists and engineers working within the field, and support collaboration to enable process improvements benefitting future cell therapy patients.

2.0

In/out of scope

Due to the large and diverse nature of the cell and gene therapy field, it was agreed that detailing every process, technology and unique aspect would not be possible. Additionally, due to current variability in processes between companies, depicting every process route would result in highly complicated (and confusing) process maps. Therefore, to enable ease of use of the detailed process maps a number of aspects were agreed to be out of scope. Below is a list of aspects that the team included in and out of scope.

In scope

- example high-level workflows describing one possible route for manufacturing cell and gene-modified cell therapies for both autologous and allogeneic formats
- maps that represent methods in use by the majority of cell therapy manufacturers today
- high-level steps for workflows, including incoming ancillary materials (e.g. buffers, media, reagents, virus) and outgoing final product
- brief explanation of the process and example technologies for each process step box (not an exhaustive list)

Out of scope

- exhaustive list of all processes, methods, and technologies in use for therapy manufacturing today
- process maps describing fully manual or fully closed and automated processes for autologous and allogeneic manufacturing
- manufacturing processes for ancillary materials, e.g. buffers, reagents, viral vectors
- clinic-related steps including administration to patient, e.g. thawing at the clinic
- in-process control (IPC) / process analytical technology (PAT) testing
- quality control sampling points – study is focused on manufacturing aspects and assume each company will test at required points as per own control strategy and regulatory requirements
- discussion of specific technologies or supply chain logistics
- detailed step-by-step methodology indicating each equipment/handling step
- timing – processing or turnaround
- variation of scale
- specific sources, cell types or therapies not covered – bone marrow, mesenchymal stem cells (MSCs), embryonic stem cells, haemopoietic stem cells (HSCs)

3.0

Overview of the process maps

Each box is a hyperlink to the full process map, with legend, in the appendices.

Autologous cell therapy	Patient treatment cycle	Viral gene editing manufacturing	Non-viral gene editing manufacturing			
Allogeneic cell therapy	Patient treatment cycle	iPSC generation and cell banking	iPSC cultivation and freezing	iPSC differentiation and filling	CAR-T cell selection, editing and expansion	CAR-T cell product

Legends

The following is a list of legends indicating what each process map depicts. These are not intended as a full description of the process, as the maps are intended to be used independently. The text below each process map box provides a brief explanation/function and some example technologies for that step. Additionally, each map contains steps above the workflow describing points for addition of buffers, raw materials, and key ancillary materials.

Appendix 1: Autologous cell therapy, patient treatment cycle

Autologous patient treatment cycle from diagnosis through administration at the clinical site. Example includes cell preparation scenarios for local and centralized manufacturing. The block flow diagram also includes links to more detailed associated workflows covering manufacturing of viral and non-viral gene-edited products (appendix 2 and appendix 3). Color-coded pathways represent different shipping/cryopreservation options for cells.

Appendix 2: Autologous cell therapy, viral gene editing manufacturing process

Model autologous workflow for manufacturing chimeric antigen receptor T cell therapies (CAR-T) using **viral** gene editing technologies. Steps include cell prep, selection, activation, genetic modification, expansion, and harvest/formulation to generate fresh and cryopreserved CAR-T cells. Final containers and labels are included as ancillary materials due to criticality of labelling during process and potential for this step to occur in manufacturing suite.

Appendix 3: Autologous cell therapy, non-viral gene editing manufacturing process

Model autologous workflow for manufacturing chimeric antigen receptor T cell therapies (CAR-T) using **non-viral** gene editing technologies. Steps include cell preparation, selection, activation, genetic modification, expansion, and harvest/formulation to generate fresh and cryopreserved CAR-T cells. Final containers and labels are included as ancillary materials due to criticality of labelling during process and potential for this step to occur in manufacturing suite.

Appendix 4: Allogeneic cell therapy, patient treatment cycle

Allogeneic patient treatment cycle from diagnosis through administration in the clinic. Associated appendix 7 indicates separate workflow detailing methods for generation of target cells via differentiation of fresh and frozen induced pluripotent stem cells (iPSCs).

Appendix 5: Allogeneic cell therapy, iPSC generation and cell banking

Generation and banking of genetically modified iPSCs from patient samples for good manufacturing practice (GMP) manufacturing. Steps include sample collection, selection, genetic modification, cloning, cell banking and bank testing. Process-flow diagram describes production of starting materials used in appendix 6.

Appendix 6: Allogeneic cell therapy, iPSC cultivation and freezing

Large-scale expansion, harvest, and cryopreservation of iPSCs from working cell banks (WCB). Appendix 5 indicates links from separate workflow detailing generation of the iPSC WCB. Process results in starting materials for iPSC differentiation into target cell types (appendix 7).

Appendix 7: Allogeneic cell therapy, iPSC differentiation and filling

Differentiation of iPSCs into target cells from WCBs of cryopreserved cells generated in workflow appendix 6. Steps cover differentiation, selection, and formulation of fresh and cryopreserved cell products for shipment to the clinical administration site. Work precedes patient treatment cycle described in appendix 4.

Appendix 8: Allogeneic cell therapy, CAR-T cell selection, editing and expansion

Manufacturing of CAR-T cell therapy using viral gene editing. Steps include cell selection, activation, genetic modification (gene knockout, delivery), and expansion. Map describes production of starting materials for cell product generation depicted in appendix 9.

Appendix 9: Allogeneic cell therapy, CAR-T cell product

Processing of cell product using starting materials produced in workflow appendix 8. Steps include selection of CAR+ TCR- cells through harvest, formulation, and cryopreservation. Final containers and labels are included as ancillary materials due to criticality of labelling.

4.0

Conclusions

Standard methodologies for generating therapeutic cell products (with or without gene modification) continue to change as the field matures. BioPhorum has identified significant gaps in publicly available information describing industrial practices for production of cell and gene-modified cell therapies. The process maps created by BioPhorum's Cell & Gene Therapy Commercialization workstream, included in this study, describe a variety of known manufacturing processes in the field of cell and gene-modified cell therapy. They are intended to enable discussion across the field and support accelerated commercialization of novel modalities.

The Commercialization workstream will continue to leverage these example process maps as a tool to identify outstanding challenges in the field and potential mitigation strategies to drive improvement and standardization in cell and gene-modified cell therapy manufacturing.

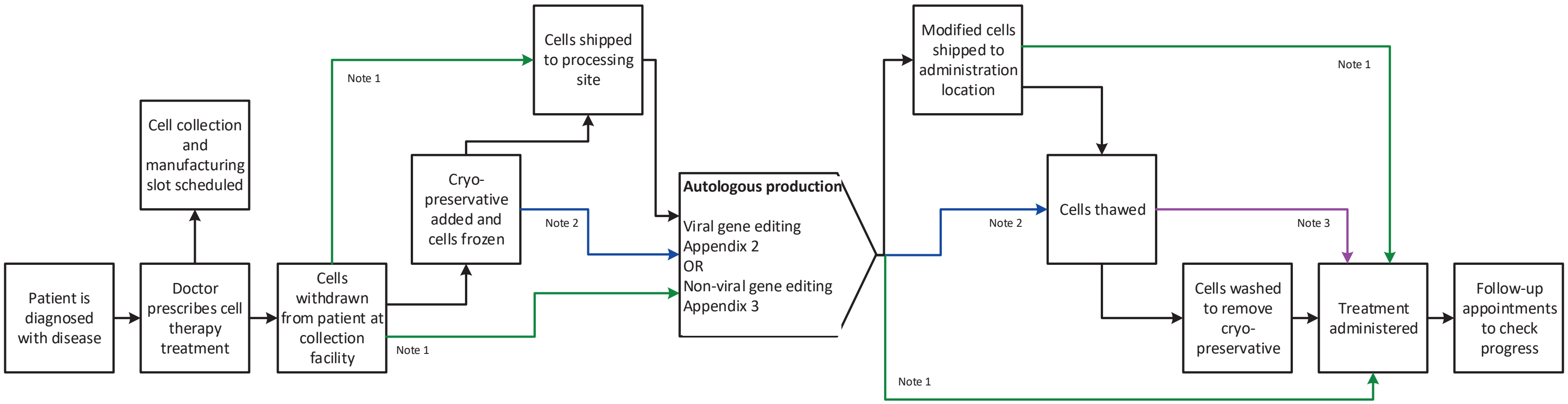
Appendices

Process maps

The following 9-page appendices contain the process maps.

These process maps are available to download and print in A3.

Appendix 1: Autologous cell therapy, patient treatment cycle



Explanation/Function	Collect apheresis or whole blood	Wash, concentrate and resuspend cells for freezing in vials or bags	Cells shipped to processing facility	Cell modification as detailed in linked referenced maps	Hospital or clinic	Thawing of cryopreserved cells	Cryopreservative replaced with final formulation solution	Single or multiple doses are administered at the hospital or clinic Patient held for observation after treatment	Patient returns to doctor to evaluate treatment effectiveness
Technology platforms	Apheresis apparatus	Tangential flow filtration or centrifugation Cryopreservative (± DMSO)	Dry ice shipper Liquid nitrogen transport dewar Track-and-trace system Temperature monitoring		Dry ice shipper Liquid nitrogen transport dewar Track-and-trace system Temperature monitoring	Waterbath or dry thawing device	Tangential flow filtration or centrifugation	Infusion of cells in blood stream Surgical implantation of cells in target tissue	Data shared with insurance company and provider of therapy to determine reimbursement

Notes

1. Represents case In which cells are not frozen. Cells processed locally or shipped without cryopreservation.
2. Represents case in which cells are processed locally and cryopreserved as part of manufacturing process.
3. Product has injectable cryopreservative.

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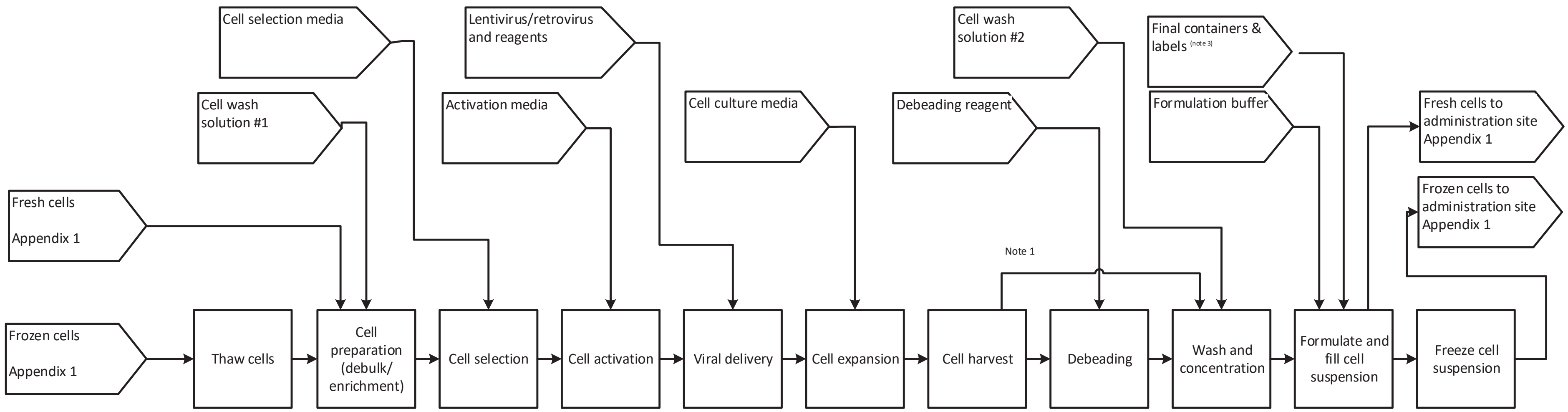
Appendix 2: Autologous cell therapy, viral gene editing manufacturing process



Explanation/Function	Thaw samples in bags/cryovials	Reduce contaminating cell types (e.g. RBCs/ platelets) from apheresis or whole blood	Select CD3+, CD8+, CD4+ T-cells from PBMCs	Stimulate T-cell growth and enhance uptake of viral material	Deliever genetic material to nucleus Washout excess virus	Expand gene modified T cells in static or agitated culture systems	Pool cell culture material and prepare to purify and formulate	Remove cell selection or activation media from cell suspension	Purify cell suspension and concentrate to the desired volume for administration	Place cell suspension in solution for administration or cryopreservation Fill, inspect, and label final container	Freeze suspension to preserve cells during shipment or storage
Technology platforms	Water bath Dry thawing device	Density gradient centrifugations	Affinity purification using magnetic or other beads	Beads coated with activators such as CD3/CD28 Non-bead technologies Interleukins	Lentivirus or retrovirus delivery system Transduction using spinoculation or reagents	Single-use bioreactors Gas permeable bags Flasks Cell culture media, cytokines, growth factors and/or serum	Depth filtration Tangential flow filtration, Centrifugation	Magnetic cell separation	Tangential flow filtration Centrifugation	Tangential flow filtration Centrifugation	Controlled rate freezer Cryo-bags Cryo-vials

Notes

1. Some processes may not include a debeading step. Cell selection or activation agents may be present in the cell product.
2. All steps are identical to the appendix 3 map apart from the red process box columns.
3. Final containers and labels added as ancillary material due to criticality of labelling during process. This may occur within a manufacture suite rather than in a separate unit.

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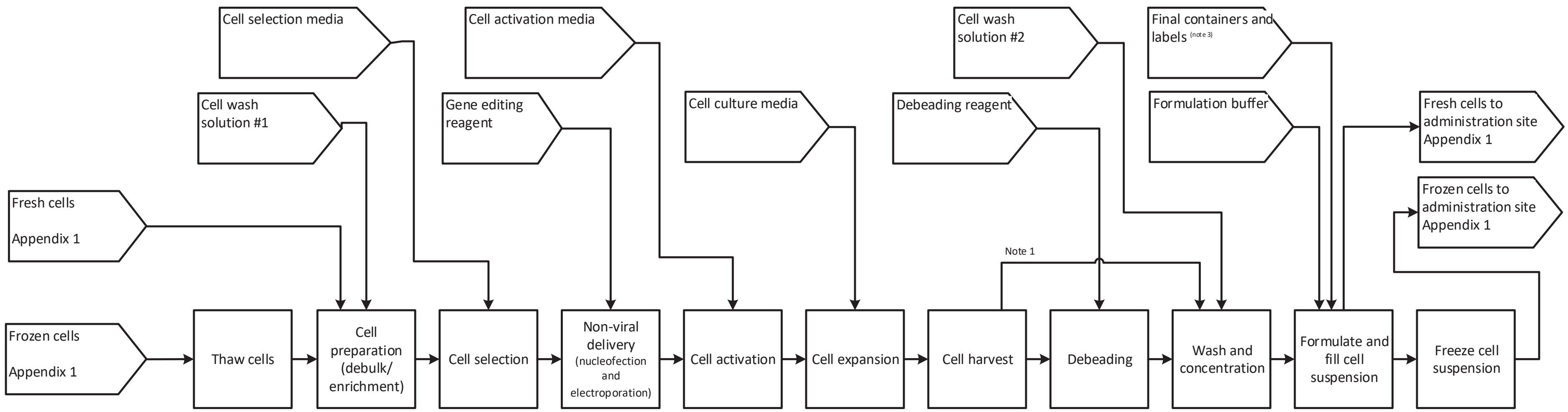
Appendix 3: Autologous cell therapy, non-viral gene editing manufacturing process



Explanation/Function	Thaw samples in bags/cryovials	Reduce contaminating cell types (e.g. RBCs/ platelets) from apheresis or whole blood	Select CD3+, CD8+, CD4+ T-cells from PBMCs	Deliever genetic material to nucleus	Stimulate T-cell growth	Expand gene modified T-cells in static or agitated culture systems	Pool cell culture material and prepare to purify and formulate the cell suspension Cell separation	Remove cell selection or activation media from cell suspension	Purify cell suspension and concentrate to the desired volume for administration	Place cell suspension in solution optimized for administration or cryopreservation. Fill final container Perform final inspection Label final container	Suspension frozen to preserve cells during shipment or storage
Technology platforms	Water bath Dry thawing device	Density gradient centrifugation Spinning membrane technologies Tangential flow filtration	Affinity purification using magnetic or other beads	Transposons (e.g. Sleeping Beauty, Piggyback) Targeted nucleases (e.g. CRISPR/Cas9, ZFN) Delivery system (e.g. electroporation or transfection reagent)	Beads with surface activators such as CD3/CD28 Non-bead technologies Interleukins	Single-use bioreactors Gas-permeable bags Flasks Cytokines, growth factors and/or serum in cell culture media	Depth filtration Tangential flow Centrifugation	Magnetic cell separation	Tangential flow filtration Centrifugation	Tangential flow filtration Centrifugation	Controlled rate freezer Cryo-bags Cryo-vials

Notes

1. Some processes may not include a debeading step. Cell selection or activation agents may be present in the cell product.
2. All steps are identical to the appendix 2 map apart from the red process box columns.
3. Final containers and labels added as ancillary material due to criticality of labelling during process. This may occur within a manufacture suite rather than in a separate unit.

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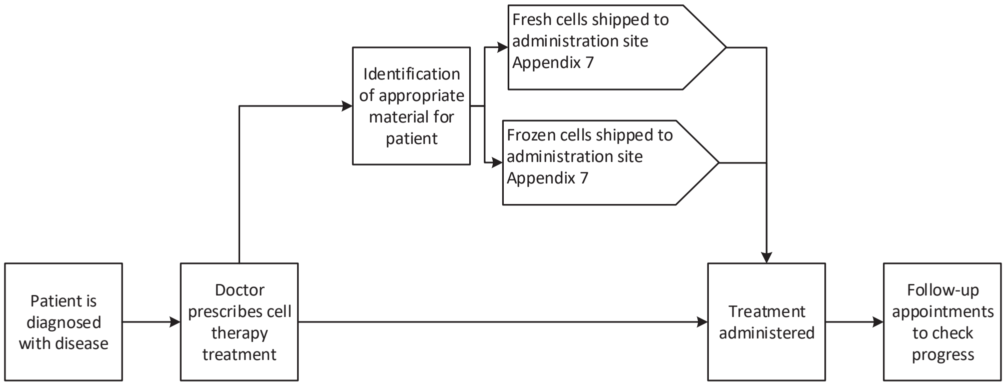
Appendix 4: Allogeneic cell therapy, patient treatment cycle



Explanation/Function	Prescription of an allogeneic cell therapy treatment	Starting cells should be a match for the patient's cell type Cell type information is transmitted to the manufacturer	Ship cells from manufacturer to administration site	Single or multiple doses are administered at the hospital or clinic Patient held for observation after treatment	Patient returns to doctor to evaluate treatment effectiveness
Technology platforms	Tests are run on patient to characterize the target cells of the therapy		Dry ice shipper Track-and-trace system Temperature monitoring	Cells thawed if frozen, warmed to body temperature Infusion of cells into blood stream Surgical implantation of cells in target tissue	Data shared with insurance company and provider of therapy to determine reimbursement

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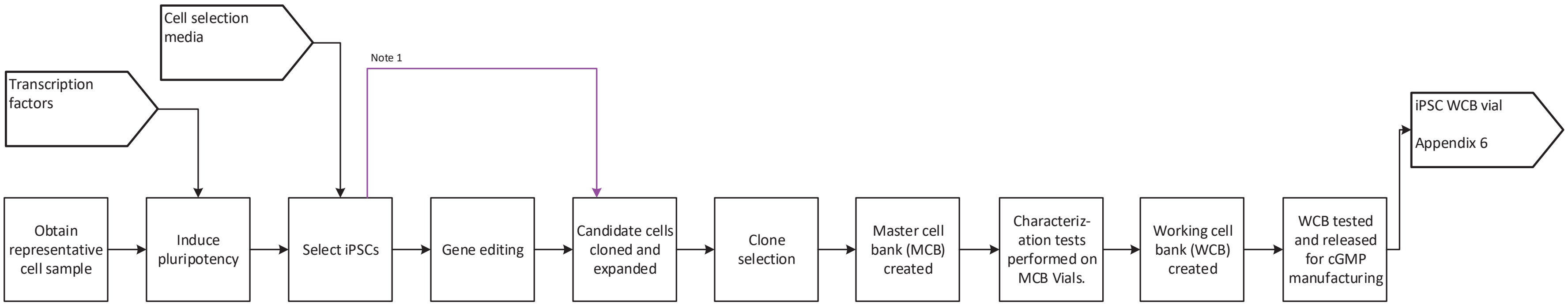
Appendix 5: Allogeneic cell therapy, iPSC generation and cell banking



	Obtain representative cell sample	Induce pluripotency	Select iPSCs	Gene editing	Candidate cells cloned and expanded	Clone selection	Master cell bank (MCB) created	Characteriz-ation tests performed on MCB Vials.	Working cell bank (WCB) created	WCB tested and released for cGMP manufacturing
Explanation/Function	Sourcing representative cells for reprogramming	Somatic cell reprogramming	Target cells are selected using surface or pluripotency markers	Genome editing to influence the histocompatibility profile of stem cells	Ensure clonality and then expand to gain sufficient cells for selection	Select cells that meet product quality criteria (e.g. growth rate, ability to differentiate into desired cell type) Vector clearance Screening	Establishes stem cell source to be used throughout the product lifecycle (e.g. 20+ years) iPSC bank will possibly be used for multiple products	Analytical testing to demonstrate consistency upon multiple scale-ups and verify absence of adventitious agents Consistent performance demonstrated – multiple expansions	New WCBs made as needed from the MCB	Demonstrates consistency upon multiple scale-ups and verifies absence of adventitious agents
Technology platforms	Fibroblasts, PBMC or CB cells	Small-scale culture equipment Example transcription factors Oct-¾, SOX2, cMyc, Kil4, Nanog and LIN-28 Efficient delivery methods for factors	Magnetic Cell Sorting Flow cytometry High Throughput clone selection equipment Manual selection	Viral or non-viral gene editing methods	Automated colony selection and liquid handler High-throughput plate-based culture system Flow cytometry ELISA	High-throughput screening platform	Small-scale culture equipment	Growth tests (out of thaw and sustained growth rates) Productivity assessments Cellular impurity characterization assays Small-scale culture equipment	Small scale culture equipment and/or bioreactor Formulation buffer ± DMSO Cryovials and/or cryobags Controlled rate freezer	Adventitious agent and genetic testing Cell performance evaluation Growth tests (out of thaw and sustained growth rates) Productivity assessments Cellular impurity characterization assays Small-scale culture equipment

Notes
1. Gene editing may not be performed in all cases (e.g. when cells are encapsulated for administration).

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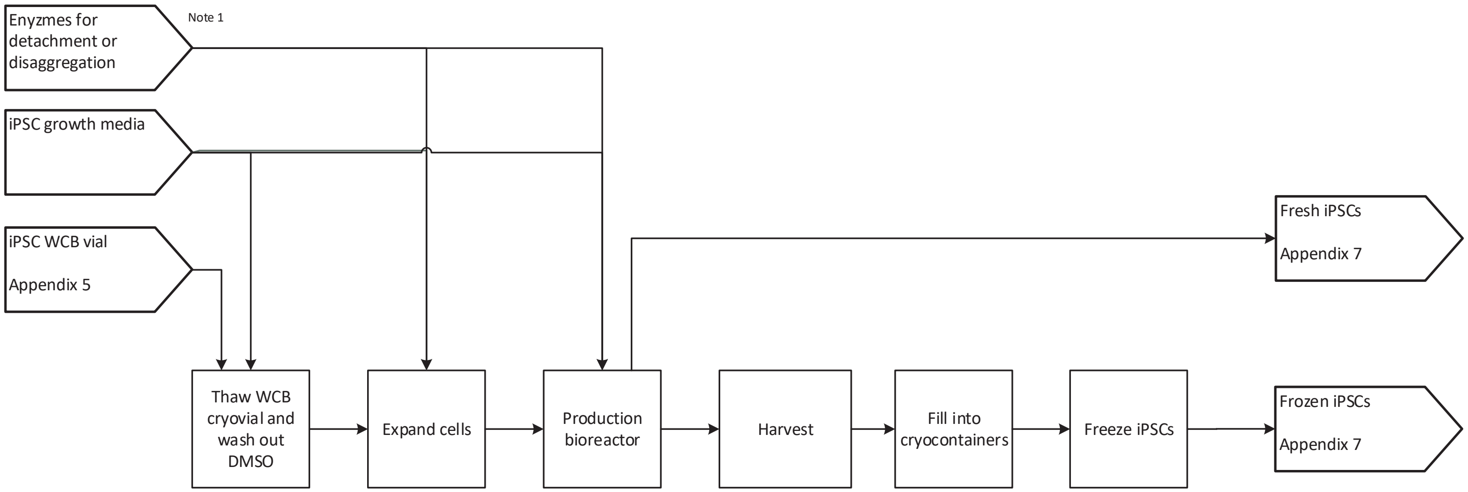
Appendix 6: Allogeneic cell therapy, iPSC cultivation and freezing



	Thaw WCB cryovial and wash out DMSO	Expand cells	Production bioreactor	Harvest	Fill into cryocontainers	Freeze iPSCs
Explanation/Function	Thaw cells, wash out DMSO and cryopreservative	Expand iPSCs in planar or suspension culture systems	Expand IPSCs to achieve target culture volume and cell density Adherent cells detached or aggregates dissociated between passages	Wash Suspension, adjust cell concentration, and add cryopreservative	Aliquot formulated culture into containers for freezing	Freeze cells to maximize viability upon thaw
Technology platforms	Waterbath or dry-thawing device Centrifuge	Adherent cell culture platform. Enyzmatic detachment of cells at each passage (if required) Suspension platform using rho-kinase inhibitor medium Controlled aggregate sizes	Suspension cultures: Single-use bioreactors, media for enzyme dissociation (if required) Adherent cultures: planar surfaces, microcarriers or hydrogels	Centrifugation	Sterile cryobag manifolds Cryovials filled in aseptic, robotic filling machine Manual filling using a pipette under laminar air flow	Controlled rate freezer

Notes
1. Not needed in every process.

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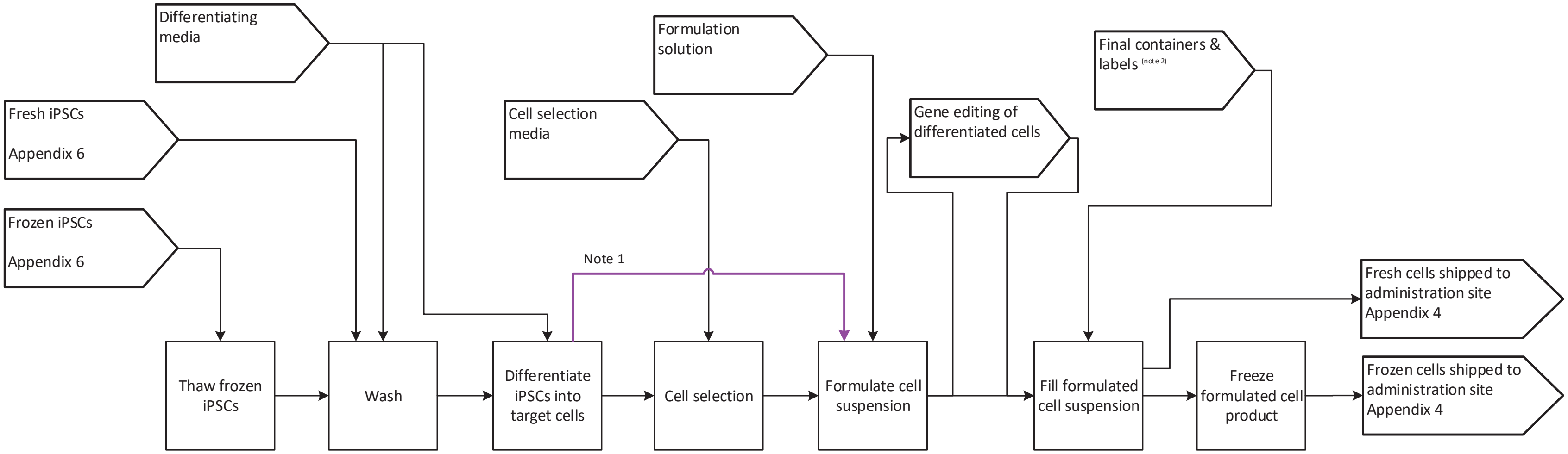
Appendix 7: Allogeneic cell therapy, iPSC differentiation and filling



	Thaw frozen iPSCs	Wash	Differentiate iPSCs into target cells	Cell selection	Formulate cell suspension	Fill formulated cell suspension	Freeze formulated cell product
Explanation/Function	Thaw iPSCs for differentiation	Wash cryoprotectant from thawed cells Media exchange to formulation for differentiation – including recovery phase in appropriate media to enable cells to recover	Introduce growth factors using protocol to differentiate cells into target cells	Remove unwanted cells or actively select desired cells	Cell suspension placed in formulation buffer which most likely contains a cryoprotectant	Fill, inspect, and label final container	Suspention frozen to preserve cells during shipment or storage
Technology platforms	Water bath Dry thawing device	Centrifugation	Single-use bioreactors Gas-permeable bags Flasks Cytokines, growth factors and/or serum in cell culture media Adherent or suspension cell culture	Magnetic cell sorting Steric cell sorting	Centrifugation	Robotic Aseptic filling machine Pre-stoppered, sterile vials Laser seals stopper Automated filling of large manifolds of sterile bags Sterile cryobags and cryovials	Controlled-rate freezing

Notes

1. Some processes may not feature a cell selection or depletion step. Culture may be sent directly to the formulation step.
2. Final containers and labels added as ancillary material due to criticality of labelling during process. This may occur within a manufacture suite rather than in a separate unit.

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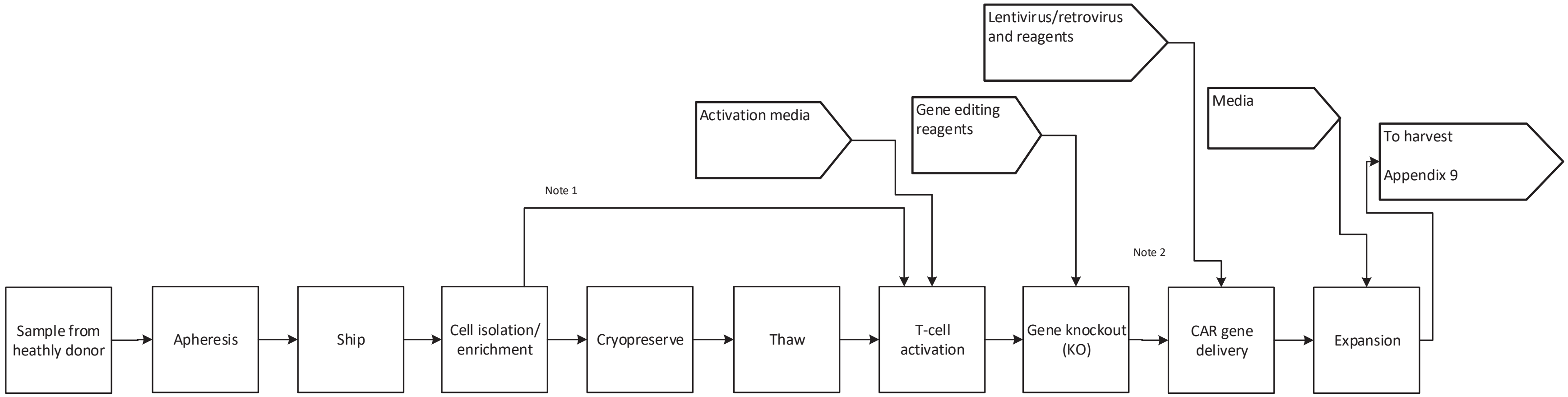
Appendix 8: Allogeneic cell therapy, CAR-T cell selection, editing and expansion



	Sample from heathly donor	Apheresis	Ship	Cell isolation/ enrichment	Cryopreserve	Thaw	T-cell activation	Gene knockout (KO)	CAR gene delivery	Expansion
Explanation/Function	Screen for the desired phenotype and function of cells (T-cell, Treg, NK, GDT)	Obtain the desired cell type Blood draw via centrifugation separation	Cells shipped to processing facility fresh, 2-8^{0}C, or frozen	Isolate desired target cell type from apheresis product	Preserve and freeze cells	Warm and thaw cells prior to further processing (option to pool multiple batches of cells)	Stimulation and proliferation of cells	To make the cells allogenic	To insert gene of Interest (GOI)	Increase cell population
Technology platforms	Collect whole blood and isolate PBMCs Ficoll for cell separation	Optia apheresis system Apheresis unit and centrifuge	Dry ice shipper Liquid nitrogen transport dewar Track-and-trace system; temperature monitoring	Affinity purification using magnetic or other beads Cell sorter (Fluorescence-activated cell sorting - FACS)	Controlled rate freezer Bags or vials	Water bath Dry thaw devices	Antibody-coated beads, bags, flasks. Soluble activation reagents/antibodies Antigen presenting cells (APC)	Delivery systems: electroporation, nucleofectors, lipid nano particles, viral, mechanical squeeze technology Technology: ZFN, TALEN, CRISPR, RNAi	AAV, lenti, retrovirus	Flasks, suspension in bags, bioreactors, perfusion

Notes

1. In some cases, freezing step may be skipped.
2. Order of gene knockout and CAR-T gene delivery may be reversed.

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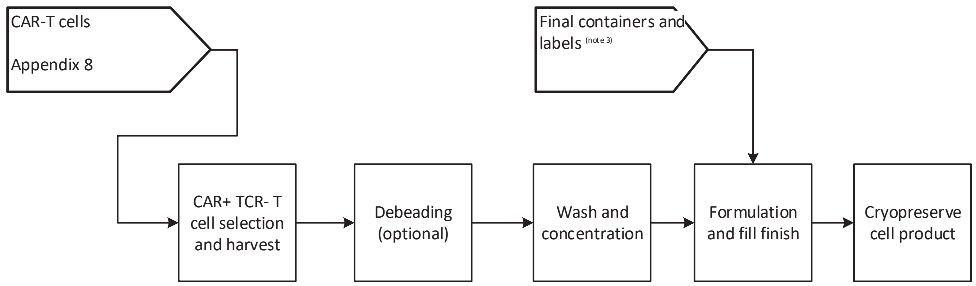
Appendix 9: Allogeneic cell therapy, CAR-T cell product



	CAR+ TCR- T cell selection and harvest	Debeading (optional)	Wash and concentration	Formulation and fill finish	Cryopreserve cell product
Explanation/Function	Obtain product purity	Remove cell selection or activation media from cell suspension	Remove culture media / Prep for formulation	Final preparation of cell product (Concentration, freeze medium, excipients) Fill final container Perform final inspection Label final container	Freeze for long term storage
Technology platforms	Affinity purification using magnetic or other beads FACS	Magnetic cell separation	Centrifugation Filtration	Vials Suitable for cryopreservation Cryobags	Controlled rate freezer Cryo-bags Cryo-vials

Notes

1. In some cases, freezing step may be skipped.
2. Order of gene knockout and CAR-T gene delivery may be reversed.
3. Final containers and labels added as ancillary material due to criticality of labelling during process. This may occur within a manufacture suite rather than in a separate unit.

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